

APTITUDE ASSESSMENT ENGINEERING DOMAIN



*Engineering & Technology Aptitude Assessment
based on Intrinsic Factors*



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A Message



Message

'Lovely Professional University is the largest single campus university in India, with more than 25,000 students, 3500 faculty and staff, offering more than 200 programmes.

The university is recognized by University Grant Commission (UGC) and is member of coveted Association of Indian Universities (AIU) and has 50+ foreign tie-ups for equitable and affordable quality education.

The university campus exhibits a rich diversity as the academic staff and students come from all the states of India and more than 26 countries in the world. Education at LPU is a holistic one, aimed at developing the intellectual and personal strengths of students. LPU has an excellent track record of launching its students onto successful careers which has been proved by our Alumni presently working in 300+ global and Indian Super Brands.

Taking a step forward in our further progress, LPU is providing an opportunity to students for identifying their true potential and strengths and be capable of choosing the most suitable career for themselves. In order to achieve academic excellence, the career that the student chooses becomes very important and taking the first right step in this direction becomes crucial, in which, the idea of 'career assessment test' will help the students in choosing the best option.

In this era, when vast varieties of career options are available to the students, it becomes all the more important to choose the best career according to their aptitude, personality and other factors and thus, be empowered to achieve the desired objective.

With the wide range of career options available with LPU whether its technical, creative, managerial etc, the student is free to choose the career of his/her choice. LPU provides ample opportunity for students to pursue and fulfill their dreams of a promising career along with providing overall personality development to them.

We wish all the best to the future leaders and expect that this initiative of LPU will properly guide them in choosing the best career and be successful in life!

The Importance of this Assessment

Discover the most appropriate Subjects Stream

We cannot find wiser words which resonate with our philosophy than the following from Mark Sullivan who headed the United States Secret Service till 2013.

"To find a career to which you are adapted by nature, and then to work hard at it, is about as near to a formula for success and happiness as the world provides. One of the fortunate aspects of this formula is that, granted the right career has been found, the hard work takes care of itself. Then hard work is not hard work at all."

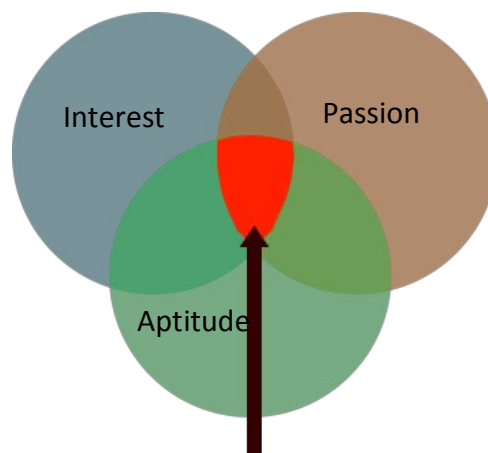
These words have been proven right time after time and the General Aptitude Assessment has been designed to assist you in selecting the most appropriate career pathways which suit not only your aptitude but also your temperament and natural propensities. The purpose is not to merely build a career but also to build a life that is fulfilling, rewarding and joyful.

WHY?

There comes a time in each student's life when he or she has to make a choice between subjects and streams which will eventually shape not only the career but also the life ahead. Very often the choices are made on the basis of factors that are extrinsic and not intrinsic.

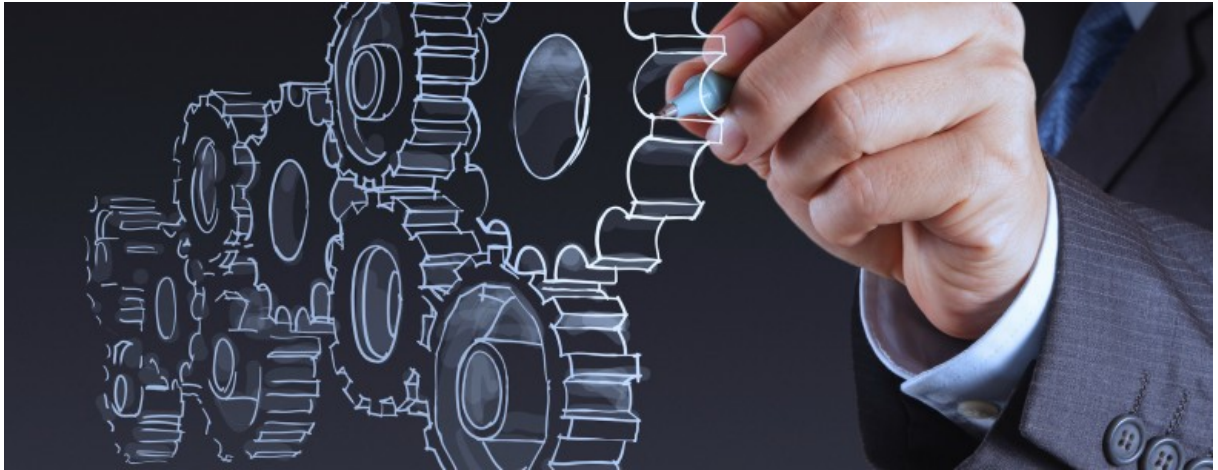
1. The choice is **imposed** by parents or relatives
2. The choice is **influenced** by current and projected trends – engineering, accountancy, medicine, management, etc.
3. The choice is **determined** by what others including peers are opting for. The suggestions are normally considered 'safe' choices.
4. The choice is **inspired** by role models

But look at the achievers in the world and it is not difficult to notice that people who have succeeded are the ones who have followed intrinsic factors:



Your Sweet Spot

Engineering & Technology Aptitude Assessment based on Intrinsic Factors



Why study Engineering & Technology?

Engineering and Technology are separate but closely related professional areas. Engineering programs often focus on theory and conceptual design while Technology programs usually focus on application and implementation.

Engineers and Technologists shape the world by turning ideas into practical reality and are involved in the design and manufacture of almost everything – from computers to aero-planes, from houses to highways, from instruments to machinery.

Course Overview

Engineering & Technology applies scientific knowledge to solve human problems and design new products. Engineers use imagination, logic, judgment, reasoning and experience to design, manufacture and operate products.

The traditional engineering & technology domain was considered to be restricted to:

1. Electrical
2. Mechanical
3. Civil

Now a wide choice is available and the career options can be viewed in terms of **Clusters** which draw upon sets of aptitudes, interests and passions.

If you have opted for the PCM Stream and are keen on making a career in Engineering & Technology you will find that there are a variety of choices before you. It is now a good time to check your aptitude for several broad areas.

Engineering & Technology Aptitude Assessment based on Intrinsic Factors

What will this report indicate?

1. The report will provide you with a bird's eye view of the current career options for students of Engineering & Technology stream.
2. It will assess your aptitude for the various rewarding career options associated with subjects of study and help you identify specific pathways to achieving your objective.

The Engineering & Technology Stream is best understood in terms of Clusters and each cluster calls for different sets of aptitude, attitude and behavioral propensities. The section on Emerging Technologies would be of special interest to you because by the time you complete the engineering graduate program several of the existing branches would have evolved into different forms.

Nature Cluster	Environmental Engineering, Biotechnology, Agricultural Engineering, Food Technology, Fermentation, Technology, Leather Technology
Chemicals Cluster	Chemical Engineering, Plastics and Polymers, Petroleum, Mining, Metallurgical
Constructive Cluster	Civil Engineering, Mechanical Engineering, Naval Construction Engineering, Aeronautical Engineering, Automotive Engineering, Electrical Engineering
Process Cluster	Process Engineering, Production Engineering, Instrumentation, Pharmaceutical Process, Process Automation
Design Cluster	Architecture, Design Engineering, Process/Product Design, Ergonomics, Urban Design
Logic Cluster	Software Engineering, Computer Engineer, Automation & Robotics, Electronics & Telecom, Instrumentation
Emerging Technologies	Sound/Audio Engineering, Ocean Engineering, Data Virtualization/Cloud Computing, Additive Manufacturing/3D Printing, Energy Systems, Nanotechnology, Simulation

Aptitude/Passion/Interest index for Engineering & Technology Courses

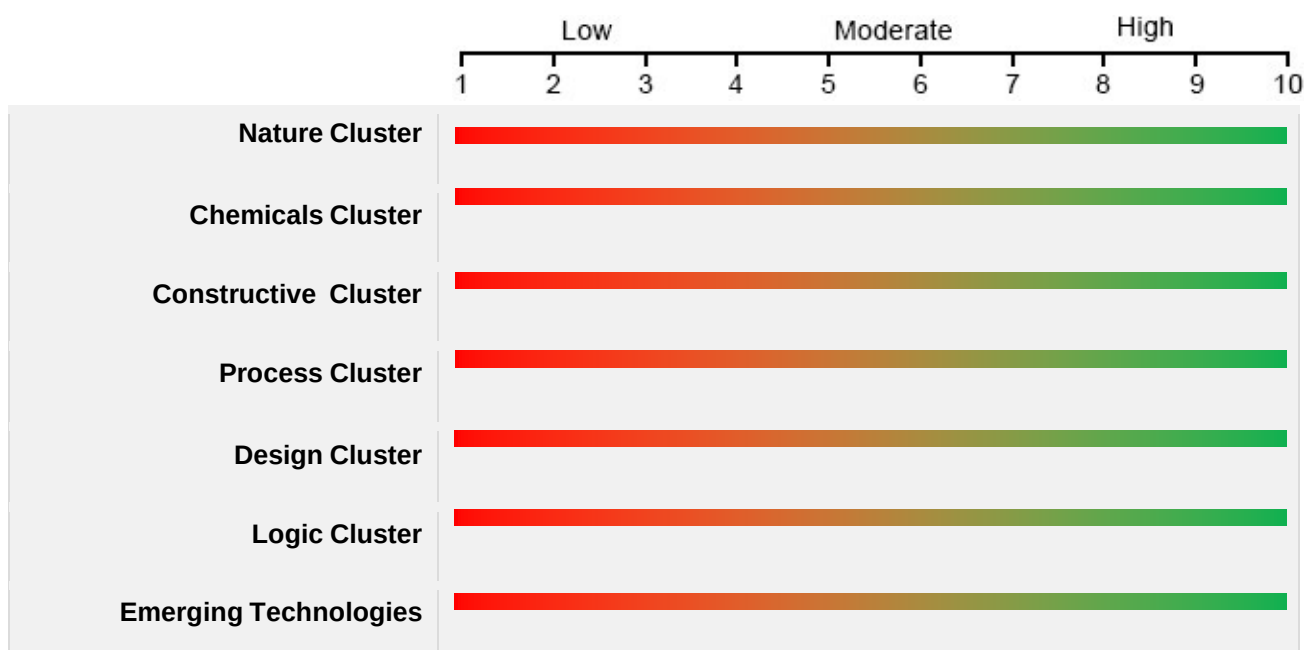
How to interpret your Engineering & Technology API Index

The graphical representation below indicates your **Aptitude + Passion + Interest Index (API)** for each of the Engineering & Technology Clusters. The assessment is based on a weighted assessment of parameters critical to each cluster.

A High API Index (towards the green zone) indicates a high intrinsic potential for excellence and achievement

A Low API Index (towards the red zone) indicates a need for realigning your effort and preferences as you may find it a bit challenging and may be not engaging enough.

A Median API Index (somewhere in the middle) indicates a 'Can DO' scenario where you may wish to invest effort and resources depending upon your ultimate career progression plan.



Your Voyage ...Our Beacon

Nature Cluster

Dear Sahil Bhardwaj

Thank you very much for taking our Second Level Aptitude Assessment Test and we sincerely hope that it would help you in making the right choice of subjects to study during the rest of your years in school within the Engineering Stream.

This would give the right thrust and direction as you prepare to enter the collegiate level of your education. The world around us is constantly evolving and new opportunities arise with each passing year. The careers that are in vogue today did not even exist a few years ago and in all probability there would be totally new or emerging careers by the time you enter college. But it is best lay the foundation NOW and be ready for the future as it emerges.

Our Assessment strongly recommends a choice for the **NATURE CLUSTER** of Engineering & Technology Careers.

This cluster involves “**nature and natural material**” – observation/involvement/sensitivity/management/analytical skills and calls for certain specific aptitude combinations. The subjects recommended for you are a combination of Physics, Chemistry and Mathematics and elective subjects depending on the options provided by your school. These subjects should give you a good foundation.

The Career Options that arise are:

After finishing School you may wish to choose a specific area to study at the graduate/post graduate level. An option you could consider is appearing for the administrative and allied services examination.

Your Voyage ...Our Beacon

Chemical Cluster

Dear Sahil Bhardwaj

Thank you very much for taking our Second Level Aptitude Assessment Test and we sincerely hope that it would help you in making the right choice of subjects to study during the rest of your years in school within the Engineering Stream.

This would give the right thrust and direction as you prepare to enter the collegiate level of your education. The world around us is constantly evolving and new opportunities arise with each passing year. The careers that are in vogue today did not even exist a few years ago and in all probability there would be totally new or emerging careers by the time you enter college. But it is best lay the foundation NOW and be ready for the future as it emerges.

Our Assessment strongly recommends a choice for the **CHEMICAL CLUSTER** of Engineering & Technology Careers.

This cluster involves “**chemicals & chemical processes**” – observation/involvement/observation/management/analytical skills and calls for certain specific aptitude combinations. The subjects recommended for you are a combination of Physics, Chemistry and Mathematics and elective subjects depending on the options provided by your school. These subjects should give you a good foundation.

The Career Options that arise are:

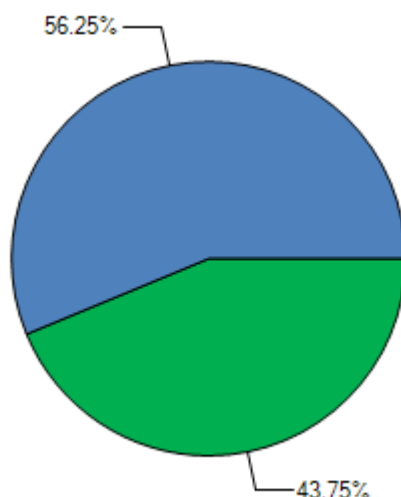
1. Chemical Engineering
2. Plastics & Polymer Engineering
3. Petroleum Technology
4. Mining Technology
5. Metallurgical Engineering

After finishing School you may wish to choose a specific area to study at the graduate/post graduate level. An option you could consider is appearing for the administrative and allied services examination.

Your ACES Personality Profile

Your personal distribution of scores on the ACES Personality Test is an indication of your unique Personality. It indicates a specific set of traits in combinations that make **'YOU'** the person that you are. The Pie Chart below is your distinctive distribution of 4 behavioral styles that make up your personality.

The biggest slice is your most dominant personality factor and the rest are your less dominant factors.



How to understand the ACES Personality Type ?

To help you understand the ACES Model of Personality definitions of the four ACES Factors considered in your report are given on the next page.

ASSERTION

The basic driving factor is ASSERTION

Driven by competition, achievement and challenge, Goal oriented and wants to be recognized by success, Ambitious, wants authority and generally resourceful, May lose interest in projects once the challenge is over and may tend to be impatient and dissatisfied with trivial matters,

Vulnerabilities: Authoritative, Argumentative, Impatient, Dominating, Inflexible, Overstepping authority

COMPLIANCE

The basic driving factor is PERFECTIONISM

Tendency to be cautious and not impulsive, Risk avoidance and preference tried and tested ways, Happy to follow procedures and rules in both personal and professional life, Preference to act in a tactful and diplomatic ways and strive for stability and orderliness,

Vulnerabilities: Excessive detail orientation, Slow decisions, Process & Procedure orientation, Indecisive

ENTHUSIASM

The basic driving factor is ENTHUSIASM

Preference to meet and be with people and quick to establish relationships, Generally optimistic, outgoing and socially skilled, Good communicator and an organizer, Preference for activities that give public recognition, appreciation and attention,

Vulnerabilities: High attention to sociability, Inattention to detail, Flamboyance, Attention span may waver

STABILITY

The basic driving factor is PEOPLE/RELATIONSHIPS

Instinctively friendly and polite, Naturally calm, patient and controlled in responses, Good and patient listener and inspires trust from others, Tendency to avoid conflict and hesitation in saying No to requests,

Vulnerabilities: Personal sensitivity, Comfort Zone preference, Resistance to change, Assertiveness

Personality Profiling (ACES)

Evaluation No.:

Name:

Personality Type	ASSERTIVE	COMPLIANT	ENTHUSIASTIC	STABLE
Emphasis	Shaping the environment by overcoming opposition to accomplish goal	Working conscientiously within existing circumstances to ensure quality	Shaping the environment by influencing or persuading others	Cooperating with other within existing circumstances to carry out tasks
Tendencies	Getting immediate results, Causing action, Accepting challenges, Questioning the status quo, Assuming authority, Crisis management, Solving problems	Concentrating on key details, Diplomatic, Checking for accuracy, Adhering to directives and standards, Thinking analytically, Using indirect approaches to conflict, Using a systematic approach to situations	Meeting people, Making a good impression, Being articulate, Creating a motivational environment, Entertaining people, Optimistic, Participating in group activities	Patience, Loyalty, Developing, specialized skills, Helping, others, Cooperating, Performing in a consistent manner, Good Listener, Creating a stable work environment
General Characteristics	Direct / Decisive, Egoistic, Risk taking, Self starter, Self motivated	Accurate, Perfectionist, Conscientious, Careful / Cautious, Systematic	Enthusiastic, Trusting, Optimistic, Persuasive, Talkative, Impulsive / Spontaneous, Emotive	Team player, Steady, Predictable, Friendly, Possessive
Preferred Environment	Power and Authority, Prestige and Challenge, Direct answers, A wide scope of operations, Opportunities for individual accomplishment, Variable roles and activities	Control over factors that affect performance, Clearly defined performance norms, Value on quality and accuracy, Reserved and business like environment, Recognition of specific skills and accomplishments	Popularity, Social recognition, Freedom of expression, Group activities outside of job, Minimal control, Favorable working conditions, Frequent coaching and counseling, like jobs that are glamorous and gives popularity.	Predictable routines, Credit for accomplishment, Good work life balance, Sincere appreciation, Identification with group, Conflict avoidance, harmony and synergy in group.

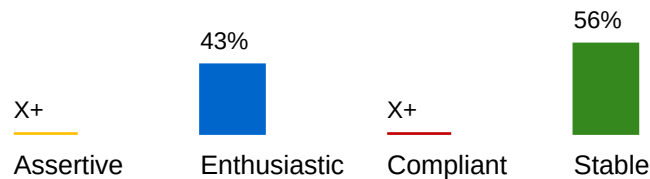
Personality Profiling (ACES)

Evaluation No.:

Name:

Personality Type	Assertive	Compliant	Enthusiastic	Stable
Value to Team	Bottom-line organizer, Maintains focus on goals, Pushes group towards decisions, High value on time, Challenges status quo, Innovative problem solver, Instinct for initiating and directing	Perspective, "Reality Check", Even tempered, Diligent, Thorough, Gathers, criticizes and tests data, Quick to see potential problems, Quality Control, Detail oriented, Instinct for organizing	Creative Problem solver, Gifted brain-stormer, Great encourager, Motivates others to achieve, Positive sense of humor, Negotiates conflicts by persuasion, Instinct for communicating	Reliable and dependable, Loyal team player, Complies with rules, Patient, Empathetic, Supportive, Peacemaker, Finds simple ways of doing things, Instinct for relationships
Possible Limitations	Oversteps authority, Argumentative, Likes to dominate, Scant regard for others, Dislikes routine, Attempts too much at once, Impatient, Can abandon ethics and take short cuts	Needs clear-cut boundaries / rules, Bound by procedures/policies/ methods, Gets bogged down in details, Prefers not to verbalize feelings, Tend to avoid arguments, Can be critical of others, Perfectionism may cause delays, Slow decision maker	More concerned with popularity and social dimension than results, Inattentive to detail, Over use of gestures and facial expressions, Wants to talk more than to listen	Resists change, Takes time to adjust to change or to take decisions, Sensitive to criticism, Can hold a grudge, Difficulty in establishing priorities, Finds it difficult to say 'No', Easily persuaded
Greatest Fear	Being taken advantage of	Criticism	Rejection	Loss of Security and Stability

Management Perspective



Preferred Style

Sahil Bhardwaj prefers roles that are team or group oriented because he enjoys working with people, coordinating programs and being a part of a team where he performs his specific role as per expectations of others. In approach, he is very cooperative and accommodating to ensure a positive team outcome. He is patient and will develop specialized skills if motivated. The tendency is to prefer a stable environment and will not appreciate abrupt changes in his environment or roles but will gradually adapt to change. He is a good listener and compliant of authority and established procedures. Can be considered reliable and dependable.

Communication expectations:

Sahil Bhardwaj expects management and others to be open and accommodating while communicating and will be very vocal in expressing his feelings and emotions. He prefers flexibility and a team approach. He may want to be noticed and hence be very communicative, expressive and use dramatic gestures. At times he can be inattentive to detail and concentrate more on the visibility and public appreciation of his work. He can be a good encourager and motivator and display a good ability to negotiate conflicts by using his influence on others.

Motivational Needs:

Sahil Bhardwaj is motivated by a need to be part of a group or team and likes a stable or steady environment. To make decisions he will prefer building consensus and adopt a democratic approach to problem solving. He will tend to put his own needs last and prefer the simplest way of doing things. He is sensitive to criticism and may tend to hold a grudge for long. He feels frustrated by loss of stability and security. Self confidence is enhanced by achievement of team goals and appreciation of his own contribution to the success of the group.

What now!



Recommended Tests:

We strongly recommend the following evaluative assessments to arrive at a more focused choice. You may wish to take these tests now or at a later stage.

Multiple Intelligence Assessment (MIA)

Professional Interpretive Assessment (PIA)

In case you require the assistance of an expert counselor please do not hesitate in contacting us.

Disclaimer

The information provided herein is confidential, privileged and protected by work-product immunity and laws related to intellectual property. The contents of this publication reflect the evaluations derived from psychometric analysis and its interpretation by experts. This publication is for informational purpose only.

The particulars of any individual's concerns and circumstances should be discussed with a qualified counselor prior to making any decisions which may affect the health and welfare of the individual. The evaluation, analysis and interpretation thereof is based on scientific studies, psychological experience or traditional usage as cited elsewhere in this publication

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Thanking you

Team LPU